Brief Resolved Unexplained Events, Apnoea and Reflux. Why the Guideline Says Do Not Test and Do Not Treat, and When Reflux Really Matters.

A seven-week-old girl is brought to the emergency department. About twenty minutes after a bottle feed she went limp in her father's arms, her lips looked dusky, and for perhaps twenty seconds she did not seem to breathe. By the time the family arrived she was feeding again and looked entirely well. Her examination is normal. Her mother mentions that she spits up after most feeds, and the first question from the admitting team is whether this was reflux, and whether she should go home on an acid blocker. It is one of the most common conversations in paediatric gastroenterology, and the evidence says that in most cases the answer to both questions is no.

The clinical question of this chapter is therefore this. Did reflux stop this baby breathing? The main idea is that reflux and pauses in breathing are both very common in infants, but they almost never happen together in time. For an infant whose event meets the criteria for lower risk, the 2016 guideline of the American Academy of Pediatrics, known as the AAP, advises against tests for reflux and against acid suppression. The evidence points instead to two other places: to swallowing, because silent aspiration is the most common positive finding after feed-related events, and to the small number of serious diagnoses, above all epilepsy, that a reflux label can hide. There is one situation in which reflux is clearly the cause of a strange-looking event, and that is Sandifer syndrome, which we will treat as reflux that you can see.

A note on sources. The definitions and recommendations come from the 2016 AAP clinical practice guideline on brief resolved unexplained events by Joel Tieder and colleagues in Pediatrics. The reflux framework comes from the 2018 joint guideline on paediatric gastro-oesophageal reflux by Rachel Rosen and colleagues for the North American Society for Pediatric Gastroenterology, Hepatology and Nutrition, known as NASPGHAN, and the European Society for Paediatric Gastroenterology, Hepatology and Nutrition, known as ESPGHAN, and from the guideline of the National Institute for Health and Care Excellence in England, known as NICE, called NG1, first published in 2015 and updated in 2019. The physiological evidence in premature infants comes from Peter and colleagues in 2002, Di Fiore and colleagues in 2005 and 2010, and Corvaglia and colleagues in 2009. The evidence on swallowing comes from two Boston studies by Daniel Duncan, Rachel Rosen and colleagues, in 2017 and 2024. The data on outcomes and risk come from a 2024 Canadian cohort study by Mahant and colleagues and from prediction rules developed by Nassr Nama and colleagues in 2022 and validated in 2025. The data on harm come from a 2012 study of ranitidine in very low birth weight infants by Terrin and colleagues, and the data on Sandifer

syndrome from reviews by Mindlina in 2020 and in Pediatric Surgery International in 2024.

Part one. From apparent life-threatening event to brief resolved unexplained event.

Until 2016 these episodes were called apparent life-threatening events, or ALTEs. The name described the fear of the observer rather than the condition of the infant, and it led to large, expensive and low-yield evaluations. The AAP replaced it with a new term, the brief resolved unexplained event, or BRUE. A BRUE is an event in an infant younger than one year, lasting less than one minute and usually less than twenty to thirty seconds, with at least one of four features: cyanosis or pallor; absent, decreased or irregular breathing; a marked change in tone, either stiffening or floppiness; or altered responsiveness. The infant must have returned to baseline, with normal vital signs and a normal examination, and the event must remain unexplained after a careful history and examination.

The last word is the most important for a gastroenterologist. If the history explains the event, it is not a BRUE. An infant who visibly regurgitates and then chokes, gags or coughs during or just after a feed has had a reflux or feeding event, and is managed as such. Redness of the face alone is not one of the BRUE colour changes. So the BRUE question, as it concerns us, is really a question about hidden reflux: reflux that nobody saw but that might have triggered the event. That is exactly the link that the physiological studies have failed to find.

Part two. Lower risk, and what the guideline advises.

The AAP defined lower-risk infants by six criteria, all of which must be met. The infant must be older than sixty days. The infant must have been born at thirty-two weeks of gestation or later and must now have a corrected gestational age of at least forty-five weeks. No cardiopulmonary resuscitation must have been given by a trained medical provider. It must be the first such event. The event must have lasted less than one minute. And there must be nothing concerning in the history or examination.

For these infants the guideline makes a set of recommendations that, in effect, protect the infant from testing. Clinicians should not investigate for gastro-oesophageal reflux, which includes pH probes, impedance studies, endoscopy, barium contrast studies, nuclear scintigraphy and ultrasound, and they should not prescribe acid suppression. Both reflux statements are moderate recommendations. The guideline also advises against blood counts, blood cultures, spinal fluid analysis, chest X-rays, echocardiograms, electroencephalography and home cardiorespiratory monitors in this group. Clinicians may observe the infant briefly, for one to four hours, with continuous pulse oximetry, may obtain a twelve-lead electrocardiogram, and may test for pertussis. And they should educate the family about BRUE, offer resources for training in cardiopulmonary resuscitation, assess social risk factors for abuse, and share decisions with the family.

How well does this work in practice? A 2024 study by Mahant and colleagues, published in BMJ Paediatrics Open, followed 1,042 infants with BRUE in eleven Canadian hospitals between 2017 and 2021. The first finding is a paradox. By the AAP criteria, 93.8 percent of these infants were classed as higher risk, so the criteria designed to spare testing spared almost no one. Nearly two thirds were admitted, and 82 percent had diagnostic tests. More than half, 52.9 percent, ended with no explanatory diagnosis. A serious underlying diagnosis was found in 7.6 percent, and the most common serious diagnosis was epilepsy or infantile spasms, in 2 percent of all infants. No invasive bacterial infections, cardiac arrhythmias or metabolic disorders were found. The most common non-serious label was gastro-oesophageal reflux, given to 25.7 percent. In other words, reflux was the label most often applied, and by definition it was not a serious diagnosis.

Part three. The mechanism: how reflux could stop breathing, and why it rarely does.

To understand why the guideline is so firm, it helps to walk through the mechanism step by step. After a feed, a small volume of refluxate, the milk and gastric fluid that flows back up the oesophagus, can reach the pharynx. In young infants most of this refluxate is not acidic, because milk buffers the gastric acid for an hour or two after a feed. At the entrance to the airway, the larynx carries chemoreceptors that detect fluid. When they are stimulated they trigger what is called the laryngeal chemoreflex: the infant swallows, the glottis closes, breathing pauses briefly, and sometimes the heart rate slows. This is a protective reflex. It keeps fluid out of the lungs, and in a healthy infant it is short and goes unnoticed. In immature infants, especially premature ones, or during an illness such as a respiratory viral infection, the reflex can be exaggerated and prolonged into an obstructive event, in which the infant tries to breathe against a closed airway.

There is an important twist. The arrow can point the other way. A 2009 study by Omari in premature infants showed that apnoea itself is associated with a fall in the pressure of the lower oesophageal sphincter, the muscular valve between oesophagus and stomach. So reflux may follow an apnoea rather than cause it, and a monitoring study that shows the two close together cannot by itself tell which came first.

This mechanism tells us what a genuine reflux-triggered event should look like. It is obstructive rather than central, it happens while the infant is awake, it happens soon after a feed, and it is often preceded by visible regurgitation, choking or coughing. A pause in breathing during sleep, far from a feed, with no effort to breathe, does not fit. The mechanism also explains why acid blockers miss the target. The trigger is fluid at the larynx, not acid. Acid suppression raises the pH of the refluxate but does not reduce the number of reflux episodes or how high they rise.

Part four. The evidence in premature infants.

The best physiological evidence comes from premature infants in neonatal units, because they have frequent apnoeas and can be studied with simultaneous recordings of breathing, heart rate, oxygen saturation and intraluminal impedance, a technique that detects the movement of fluid in the oesophagus regardless of its acidity.

In 2002, Peter and colleagues studied nineteen premature infants with apnoea of prematurity over twenty six-hour recordings. They recorded 2,039 apnoeas and 524 reflux episodes. Only 3.5 percent of apnoeas were linked to reflux reaching the pharynx. Strikingly, reflux reached the pharynx after an apnoea forty-five times and before an apnoea only twenty-six times, which fits the idea that apnoea can provoke reflux. The title of their paper summarised the result: no temporal relationship.

In 2005, Di Fiore and colleagues studied 119 premature infants and recorded 6,255 reflux episodes. Only about one percent of reflux episodes were followed by an apnoea of fifteen seconds or more, and acid reflux did not prolong apnoea. In 2010, the same group studied seventy-one premature infants and recorded 12,957 cardiorespiratory events and 4,164 reflux episodes. Fewer than three percent of events were preceded by reflux: 3.4 percent of apnoeas, 2.8 percent of drops in oxygen saturation, and 2.9 percent of episodes of slow heart rate. Reflux did not make the events longer or more severe. If anything, desaturations linked to reflux were slightly shorter.

One study points the other way, and it is worth understanding rather than dismissing. In 2009, Corvaglia and colleagues in Bologna studied twenty-six very premature infants, born at thirty-two weeks or earlier, who had recurrent apnoeas. They recorded 1,136 apnoeas and found that 154 of them, about one in seven, began within thirty seconds of a reflux episode. Apnoeas were more frequent in the thirty seconds after reflux than in the thirty seconds before, and the proportion of reflux-linked apnoeas varied widely from infant to infant. A small Italian impedance series presented in 2005 found that thirty-one of thirty-six reflux-linked apnoeas, eighty-six percent, followed non-acid reflux.

The way to put these studies together is to say that reflux-triggered events exist in a subgroup of infants, but that the subgroup is small, cannot be identified from symptoms, and is mostly triggered by non-acid refluxate. Finding these infants requires simultaneous recording of breathing and impedance with careful marking of events. And even in this subgroup, acid suppression has nothing to act on. Smaller feed volumes, paced feeding, upright positioning after feeds and simply waiting for maturation do the work. For apnoea of prematurity in general, the treatment is caffeine, not reflux medication, and the 2009 NASPGHAN consensus already stated that pathological apnoea in premature infants has no established relationship to reflux.

Part five. In the term infant with a BRUE, look at the swallow before the stomach.

If reflux is rarely the answer, what is? For events related to feeds, the most productive test is often a videofluoroscopic swallow study, in which the infant feeds barium-containing milk while a speech and language therapist and a radiologist watch the swallow on X-ray.

In 2017, Duncan and colleagues in Boston reviewed infants admitted with an apparent life-threatening event over three years, excluding infants with known neurological, cardiac or other congenital disease. The median age was forty-nine days. About twenty-nine percent had a videofluoroscopic swallow study, and seventy-three percent of those studies showed aspiration, meaning fluid entering the airway below the vocal cords, or penetration, meaning fluid entering the larynx above the cords. In twenty-six percent of infants who had both a bedside feeding assessment and a swallow study, the dysphagia was found only by the swallow study. Silent aspiration, aspiration without coughing, is common in young infants, and a bedside check misses it.

In 2024, the same group published a prospective cohort of 157 infants with BRUE seen between 2017 and 2022. In forty-two percent the event was attributed to gastro-oesophageal reflux disease, and thirty-three percent received acid suppression during follow-up. Twenty-eight percent had a swallow study, and seventy-one percent of those were abnormal. Among the sixty-six infants labelled as having reflux disease, six later had a swallow study, all six were abnormal, and the mean delay to the diagnosis of aspiration was fifty-six days. The authors' point is that the reflux label diverted these infants towards acid blockers, which do not help aspiration and may increase the risk of infection, and away from the test that would have found the problem.

A physiological study published in the Journal of Pediatrics in 2014 explains why. Ten infants with apparent life-threatening events and ten controls had manometry of the pharynx and oesophagus. Swallowing was the oesophageal event most often linked to spontaneous changes in breathing, accounting for eighty-four percent. Failure of the swallow to propagate down the oesophagus occurred in ten percent of events in affected infants and in none of the controls. The protective reflexes of the upper oesophageal sphincter were weaker, and recovery to normal aerodigestive function was slower. The authors concluded that treatment should not target gastro-oesophageal reflux; the problem is proximal, in the coordination of swallowing and breathing. A 2013 systematic review of thirty-seven studies of apparent life-threatening events by Tieder and colleagues had already concluded that none was of high quality and that testing for reflux was unnecessary. The practical message is simple. If events cluster around feeds, order a swallow study, not a pH study.

Part six. The harm of treating a reflux that did not cause the event.

If acid suppression were harmless, an empirical trial might seem reasonable. It is not harmless. In 2012, Terrin and colleagues followed 274 very low birth weight infants in Italian neonatal units, ninety-one of whom received ranitidine, a histamine-2 receptor

blocker. Infections occurred in 37.4 percent of infants given ranitidine and in 9.8 percent of those not given it, an odds ratio of 5.5. The odds of necrotising enterocolitis, a serious inflammatory disease of the premature bowel, were 6.6 times higher. Mortality was 9.9 percent with ranitidine and 1.6 percent without. This was an observational study, so it shows association rather than proof, but the direction of harm is consistent with the loss of the gastric acid barrier and changes in the gut microbiota. On the other side of the balance, randomised trials of proton pump inhibitors in infants, including the 2009 lansoprazole trial by Orenstein and colleagues, showed no improvement in symptoms over placebo, and none tested apnoea as an outcome. A treatment with a signal of harm and no demonstrated benefit is not a reasonable empirical trial.

Part seven. Risk stratification beyond the AAP criteria.

If the AAP criteria classify almost every infant as higher risk, how should we decide whom to investigate? Nama and colleagues used data from 3,283 infants with BRUE to derive what is often called the BRUE 2.0 rule, published in Hospital Pediatrics in 2022. Serious underlying diagnoses occurred in about 4.6 percent and recurrent events in about 14 percent. The predictors of a serious diagnosis were a history of a similar event, clusters of events, the need for resuscitation by a trained provider, and an abnormal past medical history. Feeding-related features were not among them. In a 2025 external validation in JAMA Pediatrics, using the 1,042 Canadian infants, the AAP higher-risk criteria had an area under the receiver operating characteristic curve of only 0.53 for a serious diagnosis. The area under the curve summarises how well a test separates two groups, where 0.5 is no better than a coin toss and 1.0 is perfect. The revised BRUE rule did modestly better, at 0.71. The practical lesson is to stratify by history: clusters, prior events, resuscitation and comorbidity should drive investigation, not reflux symptoms. Online calculators for the rule exist and can support, but not replace, clinical judgement.

Part eight. Sandifer syndrome: when reflux moves the neck.

There is one setting in which reflux is the proven cause of an alarming-looking event. Sandifer syndrome is episodic torticollis, with extension and rotation of the neck, often with arching of the back, in a child whose neurological examination is otherwise normal. NICE describes it in exactly these terms and advises specialist assessment for persistent back arching or features of Sandifer syndrome.

A 2020 review by Mindlina in the European Journal of Pediatrics collected forty-four cases from twenty-seven reports. Many children were first given neurological or psychiatric diagnoses, and some episodes were severe enough to mimic seizures. Oesophageal pH monitoring was positive in every case in which it was performed, and treating the underlying reflux led to complete or near-complete resolution of the neurological symptoms. A 2024 systematic review in Pediatric Surgery International analysed fifty-six cases.

Nineteen had an anatomical anomaly such as a hiatal hernia or malrotation, and these children needed surgery significantly more often. Of the twenty-nine children without an anomaly, twenty-three, or seventy-nine percent, improved without surgery, with a median time to resolution of one month. All cases improved with appropriate treatment of reflux.

Recognising Sandifer syndrome depends on a few features. The posturing is tied to feeds. The child is awake and responsive during the episode. There is no post-ictal phase of drowsiness afterwards. And the neurological examination between episodes is normal. The differential diagnosis is important, because the commonest serious diagnosis after a BRUE is epilepsy. Infantile spasms typically come in clusters on waking and are accompanied by a developmental plateau, and the electroencephalogram shows the chaotic pattern called hypsarrhythmia. Focal seizures show gaze deviation or automatic movements and are unrelated to feeds. Benign paroxysmal torticollis of infancy lasts hours to days, may come with pallor, vomiting or unsteadiness, and often runs in families with migraine. And structural causes in the posterior fossa or cervical spine produce a fixed tilt and an abnormal examination. A practical step is to ask the family to record an episode on video and to note its timing in relation to feeds. If doubt remains, record an electroencephalogram and an impedance study in the same admission. Labelling stiffening episodes as reflux is the error with the highest cost.

Part nine. When reflux testing does help.

Reflux monitoring has a narrow but real place after events. NICE suggests considering an oesophageal pH study, or combined pH and impedance monitoring where available, for unexplained apnoeas, for unexplained non-epileptic seizure-like events, and for suspected Sandifer syndrome. In practice the indications are recurrent events clearly tied to feeds, after the swallow has been assessed; suspected Sandifer syndrome, to confirm the association before treating or operating; and seizure-like spells with a normal electroencephalogram. The study must record impedance with pH together with oximetry and breathing channels, ideally with video and an event marker, because parents fail to report more than half of symptoms. The result should be read with a symptom association probability, a statistical measure of whether symptoms follow reflux more often than chance, where 95 percent or more supports association. Testing is not useful after a single lower-risk BRUE or for isolated apnoea of prematurity, because the pre-test probability is so low that a positive result would not mean causation. And if testing does show an association, the first steps are feed volume, pacing and positioning, with acid suppression reserved for proven acid-related disease.

Part ten. What the guidelines say, and what to do differently.

The three guidelines largely agree but differ in emphasis. On reflux testing after a lower-risk BRUE, they differ: the AAP says do not test, the 2018 joint guideline mentions impedance

only as a tool for correlating persistent symptoms with reflux, and NICE suggests considering a pH study for unexplained apnoeas. On acid suppression for events they are aligned: the AAP says do not prescribe it, the joint guideline advises against a trial of proton pump inhibitors for symptoms outside the oesophagus, and NICE does not recommend treatment for isolated symptoms. On reflux as a cause of apnoea they are aligned in substance, with NICE stating that reflux only rarely causes apnoea or apparent life-threatening events. On Sandifer syndrome they are aligned where they speak at all. Two material gaps remain. No guideline tells clinicians to assess the swallow after a feed-related event, despite how often swallow studies are abnormal. And the AAP risk criteria discriminate serious disease no better than chance.

Let me close with a few infants. A ten-week-old term infant with a first twenty-second episode of pallor and limpness and a normal examination has a lower-risk BRUE: educate the family, offer resuscitation training resources, and consider an electrocardiogram, but do not order a pH study and do not prescribe an acid blocker. A seven-week-old who coughed and choked with milk in the mouth and then became dusky has not had a BRUE but a feeding event; assess feeding and consider a swallow study. A baby born at twenty-nine weeks, now at thirty-five weeks corrected age with apnoeas, whose parents have been told it is reflux, has apnoea of prematurity; the treatment is caffeine, and acid suppression should be stopped. A four-month-old with repeated events within ten minutes of feeds needs a videofluoroscopic swallow study first, and impedance with polygraphy only if the swallow is normal. A five-month-old with neck twisting and arching during feeds and normal neurology probably has Sandifer syndrome; record a video and an impedance study, treat the reflux, and look for a hiatal hernia. And a three-month-old with clusters of stiffening on waking who carries a BRUE label does not have reflux until proven otherwise; obtain an electroencephalogram for infantile spasms.

So tomorrow in clinic, remember these points. A BRUE is brief, resolved and unexplained, and an event explained by visible reflux is not a BRUE. In a lower-risk BRUE, the AAP advises no reflux tests and no acid suppression. In premature infants, fewer than three percent of nearly thirteen thousand cardiorespiratory events followed reflux, reflux did not prolong them, and reflux reaches the pharynx more often after an apnoea than before it. A reflux-triggered subgroup exists, but it is mostly non-acid, so acid blockers have no target. After feed-related events, swallow studies are abnormal in about seven of ten infants tested, and bedside checks miss about a quarter. Ranitidine in very low birth weight infants was associated with four times more infection, sixfold higher odds of necrotising enterocolitis and a higher death rate. The AAP risk criteria predict serious disease no better than chance, and epilepsy is the leading serious diagnosis. Sandifer syndrome is reflux you can see, and four of five children without an anatomical anomaly recover without surgery. Above all, look at the swallow and the history before you blame the stomach.